EMAIL HEADER ANALYSIS AND PHISHING INVESTIGATION

Task 2 | MXToolbox Email Header Analyzer

Analysis date	10 November 2026
Tool used	MXToolbox Email Header Analyzer
Sample subject	Become a pro ■ at your aptitude skills
Sender shown	Naukri Campus Jobs <informationnc@naukri.com>

Assessment summary

The analyzed message appears to be a promotional email related to the Naukri Campus Aptitude Test (NCAT). The supplied header analysis shows successful SPF, DKIM and DMARC checks. One relay was flagged as being on a blacklist by the MXToolbox report; this is an observation that needs validation, not proof that the message is malicious.

Overall classification: **Likely legitimate based on the available evidence**. This conclusion is provisional and does not guarantee that every link or the message content is safe.

1. Objective and scope

The objective of this exercise was to inspect the supplied email header and use MXToolbox results to review sender authentication, message routing and reported blacklist status. The assessment is limited to the header/report information supplied for this task; it is not a full malware, URL or attachment analysis.

2. Email details observed

Field	Observed value
Subject	Become a pro ■ at your aptitude skills
From	Naukri Campus Jobs <informationnc@naukri.com>
Date in header	10 November 2026, 09:47:30 (+05:30)
Message theme	Promotion of the Naukri Campus Aptitude Test (NCAT)

3. Email authentication results

The supplied MXToolbox report recorded PASS for all three major email authentication mechanisms:

Mechanism	Result	Meaning
SPF	**PASS**	The sending server passed the SPF check for the domain's published policy.
DKIM	**PASS**	The DKIM signature was validated successfully.
DMARC	**PASS**	The message passed the DMARC evaluation reported by the analyzer.

Interpretation: Passing SPF, DKIM and DMARC is a positive sign that the message passed the reported domain-authentication checks. These checks do not establish that the email's content is trustworthy, that the sender account was uncompromised, or that every URL is safe.

4. Routing and blacklist observation

Relay / host shown	IP address shown	MXToolbox status
ni-central-communication-prod-service-consumer-nicampus-fcxrc27	172.10.221.151	**Reported as on a blacklist**
nmail-one03-two14-one15-one13.naukri.com	103.214.115.113	Not blacklisted

The first relay/IP is marked as blacklisted in the supplied report, while the second is marked not blacklisted. The report information available for this analysis does not establish which blacklist produced the flag, why the listing exists, or whether it was current at the time of review. The flag should therefore be independently checked against the named blacklist and current IP reputation data before drawing a conclusion.

5. Link and content observations

The message content appears to promote the Naukri Campus Aptitude Test (NCAT). The header/report includes tracked redirect links that appear to lead to Naukri Campus pages, aptitude-test content, social-media pages, an unsubscribe destination and security advice. Because the links use tracking/redirect mechanisms, the visible destination should not be treated as verified solely from the header text.

The supplied report does not provide a conclusive independent URL reputation verdict or a full inspection of page behavior. No claim is made here that every link was opened, scanned or proven safe.

6. Overall assessment

Evidence	Assessment
SPF, DKIM and DMARC all passed	Positive authentication indicators
Message promotes a Naukri Campus aptitude test	Consistent with a promotional email, but not proof of legitimacy
One relay marked as blacklisted	An anomaly requiring verification; not proof of phishing by itself
Tracked/redirect links are present	Destinations should be checked independently before clicking

Conclusion

Based on the available evidence, the email is assessed as likely legitimate rather than confirmed phishing. The successful SPF, DKIM and DMARC checks support this assessment, and the content appears promotional. However, the blacklist observation and tracked links remain items for further verification. The evidence is insufficient to label the email definitively safe or malicious.

7. Recommended next steps

1. Verify the blacklist flag for 172.10.221.151 by identifying the exact blacklist provider, listing reason and listing date.
2. Inspect each final URL destination using a reputable URL analysis service; do not click unknown links directly.
3. Confirm that the landing-page domain is an official Naukri/Naukri Campus domain and check for look-alike spellings.
4. Review the message for unexpected attachments, requests for credentials, payment, OTPs or urgent action.
5. If the message is being used as a comparison sample, retain it as a likely legitimate promotional-email example and analyze a separate known/simulated phishing sample for phishing-specific indicators.

8. Tools and limitations

Tool: MXToolbox Email Header Analyzer.
Evidence basis: The supplied email-header analysis report and its displayed results.
Limitations: This report does not include a full forensic examination of the sender's mailbox, attachment sandboxing, live URL detonation, or independent confirmation of the blacklist provider's current listing.

Privacy note for GitHub submission

Before publishing, ensure the original raw header/report and screenshots do not expose the recipient's email address, personal identifiers, message IDs, authentication tokens or unique tracking parameters. Upload only a redacted copy suitable for public viewing.

Prepared for educational cybersecurity training. Findings are limited to the evidence available in the supplied report.